

Day 5 — Identifying & Removing Delivery Bottlenecks

Activity packet for participant quads. Today's job: take the top queues from Day 4, design **bottleneck-removal experiments** with hypothesis / test / success criterion, integrate the full DP, peer-review, and ship.

Where we are in the week

The VSM exists (Day 4). Today turns its top queues into **experiments**: specific tests with specific success criteria, run on a real cadence. Then the full DP integrates and ships as the fifth and final sibling artifact.

By 16:00, every quad has the **complete sibling artifact set**: PRD / TCD / TMD / SEP / DP.

Inputs

- DP Sections 1–4
 - The top 3 queues from Day 4
 - The full sibling set (PRD / TCD / TMD / SEP)
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The Theory of Constraints — one principle

Eli Goldratt's central claim: **a system's throughput is constrained by exactly one bottleneck at a time**. Not many. One. If you optimize anything other than the current bottleneck, you don't speed up the system — you just add inventory in front of the constraint.

For software delivery: if your slowest step is "code review takes 2 days," speeding up coding doesn't help — it just creates more code waiting for review. The TPM job is to **find the current bottleneck** and remove it; **then** find the new bottleneck.

This is also why "we're working on 5 improvements simultaneously" rarely works. Pick the constraint. Move it. Then pick the next one.

Bottleneck experiment — the structure

For each top queue, design an experiment:

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### Experiment N: <short name>
**Bottleneck addressed:** <queue from VSM>

**Hypothesis:** We believe that <doing X> will
<result>
because <reasoning>.

**Test:** <what we'll do, for how long, on what
scope>

**Success criterion:** <measurable outcome that
decides
adopt/reject>
- Baseline: <current measurement>
- Target: <improved measurement>
- Window: <duration>

**What could go wrong:**
<2–3 risks of running this experiment>

**Who runs it:** <named person>
**By when:** <date>

**Decision after:**
- If success criterion is met: <adopt / scale /
institutionalize>
- If not met: <abandon / iterate / escalate>
```

The experiment **must** have a success criterion that's measurable. "We'll see if it feels better" is not an experiment — it's a vibe.

Activity 1 — Design 3 Experiments

Format: Quad • 35 min • Block 1

Purpose

Convert the top 3 queues from Day 4 into 3 specific experiments.

Quad protocol

1. **Re-read your top 3 queues** (5 min) from DP Section 4.
2. **For each, draft an experiment** (25 min) using the template above. Commit to:
 - A specific change (not "improve code review" — "auto-assign reviewers based on path")
 - A duration (typically 2–4 sprints)
 - A success criterion with baseline, target, and window
3. **Sanity-check** (5 min). For each: would your team actually try this in the next 60 days? If not, the experiment is too ambitious or too vague.

Worked example — FieldPulse

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### Experiment 1: Auto-assign code reviewers

**Bottleneck:** Build → Code review queue (1.5 days
average wait).

**Hypothesis:** We believe that auto-assigning code
reviewers
based on file paths (using CODEOWNERS) will reduce
time-to-first-
review from 1.5 days to  $\leq 4$  hours, because the
current 1.5-day
wait is mostly "who's going to review this" decision
time.

**Test:** Configure CODEOWNERS for the reconcile
module; run
for 4 sprints (8 weeks); compare time-to-first-
review and time-
to-merge against the 8 sprints prior.

**Success criterion:**
- Baseline: time-to-first-review p50 = 1.5 days, p90
= 3 days
- Target: p50  $\leq 4$  hours, p90  $\leq 1$  day, over the 4
sprints
- Window: 8 weeks; min sample 30 PRs

**What could go wrong:**
- Reviewers feel burdened by constant assignment →
measure complaints / opt-outs
- Auto-assigned reviewer is on PTO → fall-back to
next-in-list
- One reviewer becomes overloaded → measure per-
reviewer load

**Who runs it:** <eng lead> with <TPM> support
**By when:** Pilot starts next sprint (2026-05-13)

**Decision after:**
- Met: institutionalize CODEOWNERS for the whole
codebase
- Not met: examine why; possibly try paired-review
during dev
```

Activity 2 — DP Integration Pass

Format: Quad • 40 min • Block 2

Purpose

Read the full DP top to bottom. Fix incoherences. Lock the version that goes into Friday's review.

The integration checklist

Check	Pass criterion
Section 1 outcomes ↔ Section 2 backlog	Every output in Section 2 maps to an outcome in Section 1
Section 2 backlog ↔ Section 3 tracking	Every leading indicator in Section 3 corresponds to backlog items
Section 3 tracking ↔ Section 4 VSM	Cycle-time leading indicators are consistent with VSM measurements
Section 4 queues ↔ Section 5 experiments	Every top queue has at least one experiment
References to PRD/TCD/TMD/SEP	DP cites prior artifacts by section
No fortune-cookie prose	Specific to this feature

Quad protocol

1. **Solo read-through** (15 min). Each member reads Sections 1–5; marks issues in margins.
2. **Pool issues** (10 min). De-dupe.
3. **Fix in priority order** (15 min). Coherence first, then prose.

Activity 3 — Cross-Review with Friday

Rubric

Format: Quad-pair • 40 min • Block 3

Purpose

Cross-review the full DP with another quad using the Friday rubric.

The Friday rubric (DP)

Dimension	Weight	Exemplary
Outcome clarity	20%	Outcomes user-visible; tied to NS / Tier Sheet
ADO discipline	15%	Hierarchy, fields, tags, iterations populated; queries work
Output ↔ outcome pairing	20%	Each output has outcome AND leading indicator
Value stream realism	20%	Lead times measured (or honestly estimated); queues named
Bottleneck experiments	15%	Each has hypothesis, test, success criterion
Integration with prior artifacts	10%	DP references PRD/TCD/TMD/SEP appropriately

Score 0–4 per dimension. Apply weights for total /4.0. Same convention as prior weeks.

Cross-review protocol (40 min)

1. **Pair quads** (5 min). Same pair as prior weeks if possible.
2. **20 min cross-read** with the rubric.
3. **10 min author-quad responds** to feedback.
4. **5 min final score** captured.

Activity 4 — Sign-Off + Bridge to Week 8

Format: Quad • 45 min + Wrap • Block 4

Purpose

Sign off the DP. Bridge to Week 8's capstone work.

Sign-off

The quad declares the DP **Status: Approved** (or "Approved with gaps") — same convention as prior weeks. The DP joins PRD / TCD / TMD / SEP as the fifth sibling artifact.

Bridge to Week 8 — what the capstone needs

Week 8 is the capstone. Every quad works the same subject:

Holocron, an enterprise string-management problem. You receive a **problem brief** — the business situation, the facts, the stakeholders, the constraints — and build the artifact set from it. The requirements are yours to write; nothing is pre-specified. The week is anchored on the Week-8 topic: **AI Spec Development**.

What carries over to Week 8:

- The artifact templates (PRD/TCD/TMD/SEP/DP) — used in shorter form
- The AI provenance discipline
- The negotiation / outcome / tracking muscles
- The cumulative AI-validation log

What's new in Week 8:

- **AI Spec Development** — using AI to draft technical specs that integrate with the artifact set
- The **Holocron problem brief** — a real enterprise problem, handed to you cold
- **Final artifact presentations** Friday

Quad protocol — Week 8 readiness check (30 min)

1. **Read the Holocron problem brief** (15 min). Take `holocron-problem-brief.md` and read it end to end as a quad. Do not start solving. Instead capture:
 - The problem in one sentence, in your own words
 - Who you think the primary user is — and who else the brief names
 - The three facts you find most consequential
 - The first three questions you would ask the business
2. **Pre-flight** (10 min). Which stakeholders would you interview first? Which open questions in the brief must be resolved before you can scope?

3. **First-pass scope instinct** (5 min). Without committing: what feels like release one? Bring it Monday — Day 1 tests it.

Final wrap (last 15 min)

Each quad shares:

- The **bottleneck experiment** they're most excited to run
- The **single sharpest insight** the DP added beyond the rest of the artifact set
- Their **one-sentence read of the Holocron problem** + the first question they'd ask the business

End-of-week (Week 7) checkpoint

Each quad ships:

- ✓ **Full DP** with Sections 1–5
- ✓ Section 1 Outcome map
- ✓ Section 2 Loaded ADO backlog (or paper equivalent) with field discipline
- ✓ Section 3 Tracking plan (output / outcome / leading indicator triples + cadences)
- ✓ Section 4 Value stream map with PT / LT / flow efficiency
- ✓ Section 5 Bottleneck removal experiments (hypothesis / test / success criterion)
- ✓ AI provenance log entries
- ✓ Cross-reviewed + signed off
- ✓ **Holocron problem brief read**; one-sentence problem + opening questions captured

The DP joins the PRD / TCD / TMD / SEP. Together they are the **complete sibling artifact set** built across Weeks 3–7 — the input to the Week-8 capstone.